

Mortality Among Irish Academics: A Logistic Regression Analysis

Nikol Staykova

1 Introduction

This study investigates mortality among Irish academics over a ten-year period. The dataset comprises 1,735 academics across four job grades: post-docs, lecturers, senior lecturers, and professors. There are 10 variables in total, one identifier variable (id), one outcome variable (died), four categorical variables (jobgrade, sex, smoker, died) and four continuous variables (age, sbp, dbp, chol, bmi). The primary objective of this analysis is to develop a logistic regression model to determine whether all-cause mortality is associated with a set of key risk factors.

Variables observed in the data set

Outcome Variable:

Died- Records whether the academic died over the 10-year follow-up period.

Potential Factor Variables:

Sex- The biological sex of the patient, male or female.

Age- The age of the patient recorded in years and months(A float).

Job Grade- The academic's job grade at the start of the study. Takes four values: post-doc, lecturer, senior lecturer, and professor.

Cholesterol - The serum cholesterol level of the academic, measured in mmol/L.

Smoker- Smoking status of the academic, recorded as yes or no.

BMI- Body mass index of the academic.

SBP- Systolic blood pressure, the pressure in the arteries when the heart beats.

DBP- Diastolic blood pressure, the pressure in the arteries between heartbeats.

2 Inspection of the Variable Distributions

Prior to developing the prediction model, it is important to understand the characteristics of the study population. The continuous variables were summarized using the means (**Table 1**) and the categorical ones using the frequencies (**Table 2**).

Age at study entry ranges from 40 to 64 years, with a mean of approximately 52 years. This indicates that the cohort consists primarily of middle-aged to older academics.

Systolic blood pressure ranges from 87mmHg to 280mmHg, with a mean value of approximately 137 mmHg. This is elevated compared to the normal adult range of 120 mmHg, suggesting that on average, this academic population may have higher than

optimal systolic blood pressure.

Diastolic blood pressure ranges from 36mmHg to 155mmHg and has a mean of approximately 85 mmHg, which is also slightly elevated relative to the normal range of less than 80 mmHg.

Cholesterol levels averages 5.10 mmol/L, which falls broadly within the normal range for adults.

BMI ranges from 16.2 kg/m² to 37.5 kg/m² with a mean of 25 kg/m². This value sits at the upper boundary of the healthy weight range and the lower boundary of the overweight category.

Job Grade: The majority of the group are lecturers, making up 68.13% of the study population. Senior lecturers are 16.83%, while professors make up 10.09%. Post-docs are the smallest group, representing just under 5% of academics in the study.

Mortality Outcome: A total of 183 academics died during the study period, representing 10.55% of the cohort, while the majority (89.45%) survived. This outcome distribution shows that mortality affects approximately one in ten participants.

Smoking Status: The majority of the participants reported being non smokers with 63.8%.

Sex: The cohort was predominantly male, with 1,007 male academics (58.0%) compared to 728 females (42.0%).

Table 1: Summary Statistics of Continuous Variables

Variable	Mean	Min	Max
Age (years)	51.83	40	64
Systolic Blood Pressure (mmHg)	136.99	87	280
Diastolic Blood Pressure (mmHg)	84.54	36	155
BMI (kg/m ²)	24.78	16.26	37.50

Variable	Category	Frequency	Percent (%)
Job Grade	Post-Doc	86	4.96
	Lecturer	1,182	68.13
	Senior Lecturer	292	16.83
	Professor	175	10.09
Died	Survived	1,552	89.45
	Died	183	10.55
Smoker	No	1,107	63.80
	Yes	628	36.20
Sex	Female	728	41.96
	Male	1,007	58.04

Table 2: Frequency Distribution of Categorical Variables

3 Inspection of the Variables' Relationship with the Outcome

The relationship between each variable and mortality was examined graphically, as shown in **Figures 1 to 8**. For continuous variables, we reviewed smoothed fractional polynomial plots. Some plots suggested non-linear patterns, but the confidence intervals were very wide, particularly at the extremes of the data. This indicates that there is not enough evidence to confirm non-linearity, and we cannot rule out a linear relationship. Therefore, all continuous variables will be included in the logistic regression model in their original linear form.

For categorical variables, bar plots showed variation in mortality across groups.

Job grade: Mortality increased with higher grades—post-docs (5.8%), lecturers (8.7%), senior lecturers (17.1%), and professors (14.3%)—likely reflecting the older age of staff in senior positions.

Sex: Males had a higher mortality rate (14.5%) than females (5.1%), suggesting sex may be an important predictor.

Smoking status: Smokers had a slightly higher mortality rate (13.1%) than non-smokers (9.1%), indicating a slight increase.

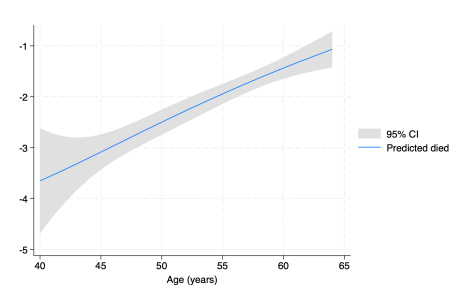


Figure 1: Relationship Between Age and Mortality

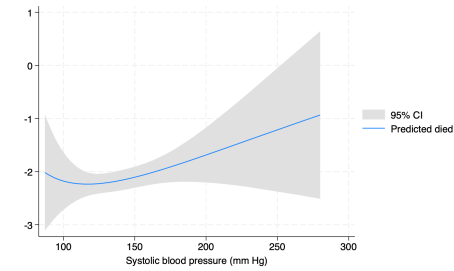


Figure 2: Relationship Between Systolic Blood Pressure and Mortality

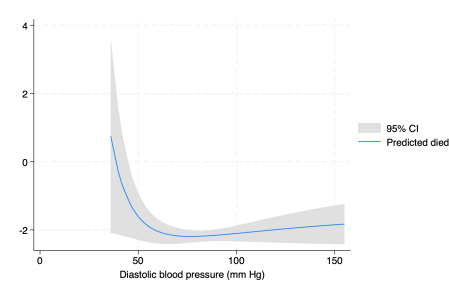


Figure 3: Relationship Between Diastolic Blood Pressure and Mortality

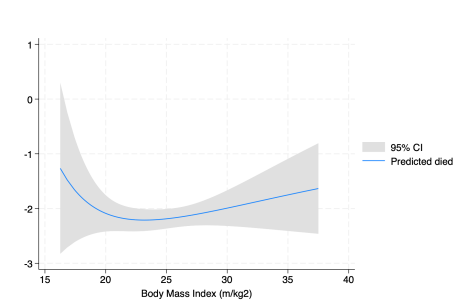


Figure 4: Relationship Between BMI and Mortality

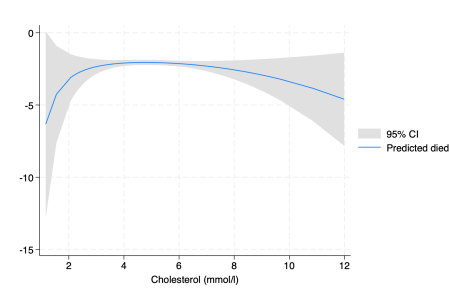


Figure 5: Relationship Between Cholesterol and Mortality

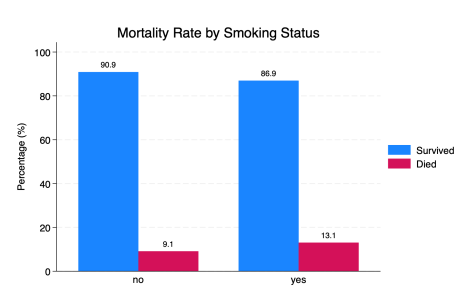


Figure 6: Mortality Rate by Smoking Status

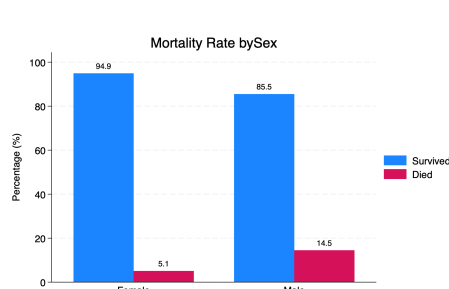


Figure 7: Mortality Rate by Sex

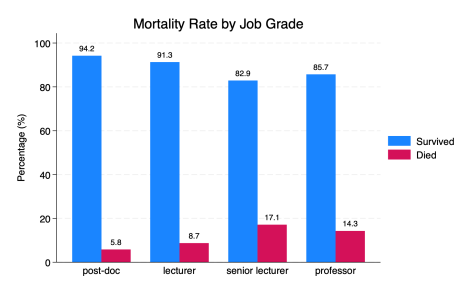


Figure 8: Mortality Rate by Job Grade

4 Model Building and Interpretation

Before fitting the model, four new variables were created by dividing the original continuous variables—age, systolic blood pressure, diastolic blood pressure, and cholesterol—by 10. This transformation ensures that the odds ratios reflect a more meaningful 10-unit increase in each variable, rather than a one-unit increase, which would result in odds ratios very close to 1 and be difficult to interpret. The full model containing all variables was fitted **Table 4** and found to be statistically significant ($p < 0.0001$), indicating that the predictors collectively explain a meaningful amount of the variation in mortality. The interpretations for each individual predictor are as following.

Significant Factors:

Age: For every 10-year increase in age, an academic has approximately 2.79 times higher odds of dying, adjusting for all other predictors. The p-value is highly significant ($p < 0.001$), confirming that age is an independent predictor of mortality.

Sex: Male academics have a 201% increase in the odds of dying compared to female academics, keeping all other predictors fixed. The p-value is highly significant ($p < 0.001$).

Smoking Status: Smokers have a 46.8% increase in the odds of dying compared to non-smokers, keeping all other predictors fixed. The p-value is significant ($p = 0.022$).

Non-Significant Factors:

Job Grade: Compared to post-docs, lecturers have a 38.5% higher chance of dying. Senior lecturers have a 66.5% higher chance, and professors have a 7.3% higher chance, with all other factors held constant. However, none of these differences are statistically meaningful ($p = 0.500$, $p = 0.311$, $p = 0.894$, respectively). This means there is no proof that job grade independently affects mortality when you consider other factors.

Systolic Blood Pressure: For every 10 mmHg increase in systolic blood pressure, an academic has approximately 3% higher odds of dying, adjusting for all other predictors. The p-value is not significant ($p = 0.568$).

Diastolic Blood Pressure: For every 10 mmHg increase in diastolic blood pressure, an academic has approximately 6.3% lower odds of dying, adjusting for all other predictors. The p-value is not significant ($p = 0.421$), suggesting that DBP does not have an independent contribution to mortality.

Cholesterol: For every 10 mmol/L increase in cholesterol, an academic has approximately 22.1% lower odds of dying, adjusting for all other predictors. The p-value is not significant ($p = 0.723$).

BMI: For every one unit increase in BMI, an academic has approximately 1.3% higher

odds of dying, adjusting for all other predictors. The p-value is not significant ($p = 0.644$).

Table 3: Full Logistic Regression Model: Predictors of Mortality in Irish Academics

Variable	Category	OR	95% CI	p-value
Job Grade	Post-Doc (ref.)	1.00	—	—
	Lecturer	1.39	0.54–3.57	0.5
	Senior Lecturer	1.67	0.62–4.46	0.31
	Professor	1.07	0.38–3.03	0.89
Age (per 10 years)		2.79	2.09–3.72	<0.001
Sex	Female (ref.)	1.00	—	—
	Male	3.01	2.03–4.45	<0.001
Smoker	No (ref.)	1.00	—	—
	Yes	1.47	1.06–2.04	0.022
SBP (per 10 mmHg)		1.03	0.93–1.14	0.568
DBP (per 10 mmHg)		0.94	0.80–1.10	0.421
Cholesterol (per 10 mmol/L)		0.78	0.20–3.11	0.723
BMI (per kg/m ²)		1.01	0.96–1.07	0.644

Model statistics: LR $\chi^2(10) = 115.91$, $p < 0.001$, Pseudo $R^2 = 0.0991$, $N = 1,735$

OR = odds ratio; CI = confidence interval; ref. = reference category. Bold entries are statistically significant at the 0.05 level.

Predicted probabilities were computed for a hypothetical male, nonsmoking academic with all continuous variables held at their mean values (age = 51.8 years, mean systolic and diastolic blood pressure, cholesterol, and BMI) in order to further investigate the relationship between job grade and mortality. According to **Table 4** and **Graph 9**, the predicted probabilities of death were 5.2% for post-docs, 7.1% for lecturers, 8.4% for senior lecturers, and 5.6% for professors. The differences between job grades are negligible, despite the apparent slight increase in predicted mortality from post-doc to senior lecturer. This is in line with the findings of the logistic regression, which showed that after controlling for all other variables, job grade was not a statistically significant predictor of mortality.

Job Grade	Predicted Probability of Death
Post-Doc	0.0521
Lecturer	0.0708
Senior Lecturer	0.0839
Professor	0.0557

Table 4: Predicted Probability of Death by Job Grade at Mean Covariate Values

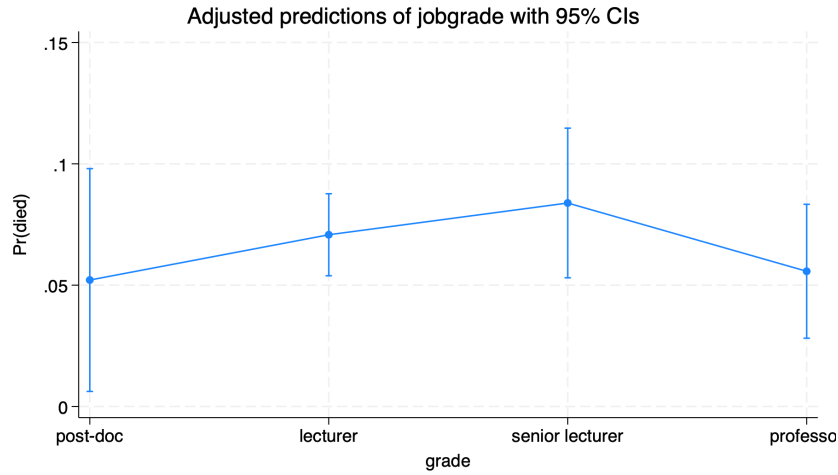


Figure 9: Predicted Probability of Death by Job Grade at Mean Covariate Values(Graph 9)

5 Internal Evaluation of the Model

5.1 Calibration

To assess the calibration of the model, the Hosmer–Lemeshow goodness-of-fit test was performed. As shown in **Table 5**, the observed and expected numbers of deaths across the ten risk groups are generally close, indicating that the model predicts actual outcomes well. This is supported by a p-value of 0.3176, which is well above the 0.05 threshold, meaning we fail to reject the null hypothesis that the model fits the data adequately.

To further evaluate internal calibration, a calibration plot was generated (**Graph 10**). The plot displays observed versus predicted probabilities across risk groups. The prediction line closely follows the reference line, particularly in the lower risk groups. This is reinforced by an observed-to-expected (O:E) ratio of 1.000, a calibration-in-the-large (CITL) of -0.000 , and a calibration slope of 1.000. Overall, these findings suggest the model is well calibrated.

Group	Prob	Obs Deaths	Exp Deaths
1	0.0238	2	3.2
2	0.0364	3	5.3
3	0.0477	11	7.3
4	0.0633	4	9.6
5	0.0797	14	12.3
6	0.1004	18	15.5
7	0.1298	25	19.9
8	0.1682	25	25.6
9	0.2270	32	34.3
10	0.4646	49	50.1
Total		183	183.1

Table 5: Hosmer-Lemeshow Goodness of Fit Table for the Full Model
Hosmer-Lemeshow $\chi^2(8) = 9.30$, $p = 0.3176$

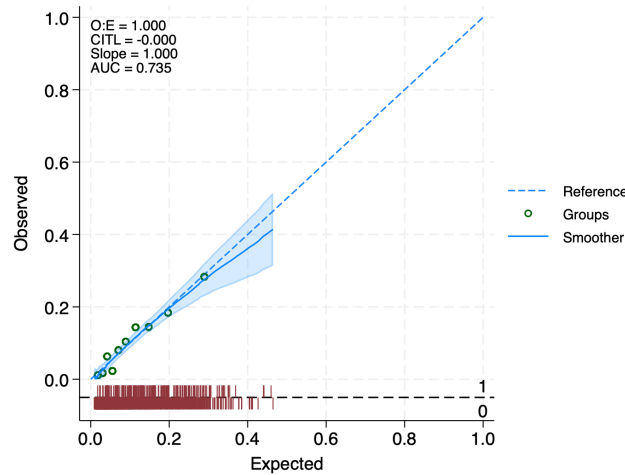


Figure 10: Prediction Model Internal Calibration Plot

5.2 Discrimination

The discriminative ability of the logistic regression model was assessed using the Receiver Operating Characteristic curve plot and a simple box plot (**Figure 11**, **Figure 12**). The model achieved an AUC of 0.7354, indicating acceptable to good discrimination. This means that when randomly selecting one academic who died and one who survived, the model correctly identifies the higher risk academic 73.54% of the time. According to standard benchmarks, an AUC between 0.70 and 0.80 is considered acceptable discrimination.

The box plot further supports this, showing that academics who died tended to have higher predicted probabilities compared to those who survived, although there is considerable overlap between the two groups, which is consistent with the moderate level of

discrimination indicated by the AUC. Combining the results, the model demonstrates good calibration and acceptable discrimination, suggesting it is a reasonably reliable tool for predicting mortality.

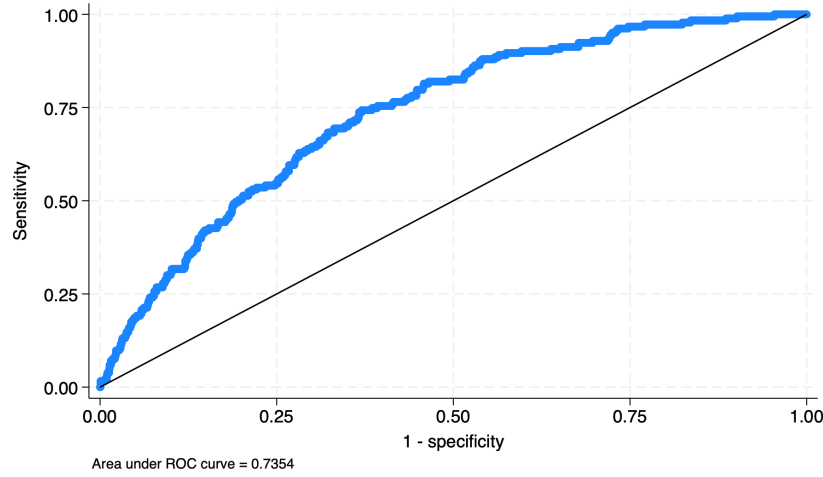


Figure 11: Receiver Operating Characteristic Curve Plot (Internal Discrimination plot)

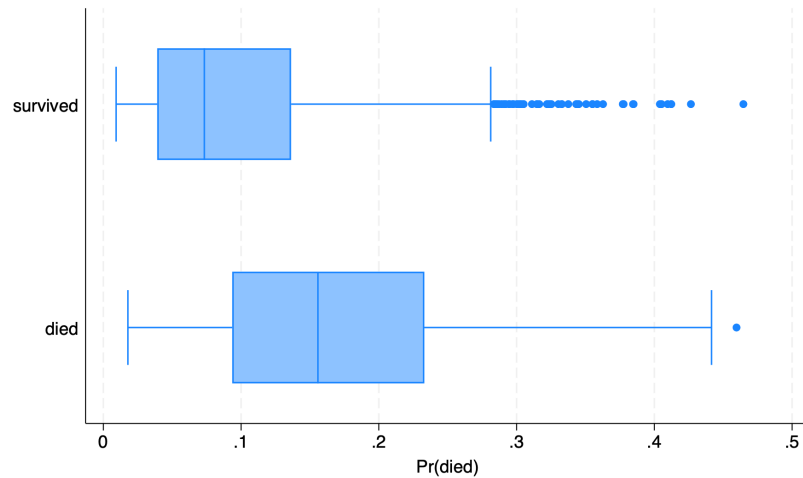


Figure 12: Predicted Probability of Short-Term Mortality by Status

5.3 Cross Validation

Ten-fold cross-validation was used to determine whether the model's calibration and discrimination results are trustworthy and not just the result of overfitting. Nine folds were used to train the model, and the remaining fold was used for testing. This process was repeated ten times. The root mean square error values were consistent across all folds, ranging from 0.256 to 0.324 (**Table 6**), with a mean RMSE of 0.299. The reliability of the calibration and discrimination results mentioned above is confirmed by this consistency, which shows that the model performs similarly on unseen data.

Fold	RMSE
1	0.2825
2	0.3072
3	0.2950
4	0.3024
5	0.3059
6	0.3245
7	0.2956
8	0.2562
9	0.3016
10	0.3159
Mean	0.2987

Table 6: 10-Fold Cross-Validation RMSE for the Full Model

6 Outlier Detection

To identify outliers in the logistic regression model, both Pearson and deviance residuals were examined. For binary outcome data the ± 2 threshold commonly used in linear regression is not appropriate, as the residuals are not normally distributed. Therefore, rather than applying a fixed threshold, the top 10 observations with the largest absolute values of each residual measure were identified and plotted. These observations are highlighted in red (**Figure 13**, **Figure 14**). Notably, the same 10 observations (IDs: 10576, 7861, 10813, 22546, 16108, 21249, 7108, 12372, 9906 and 23167) appeared in the top 10 of both residual measures, with all deviance residuals exceeding 2 and Pearson residuals ranging from 4.78 to 7.42. The consistency across both residual measures provides strong evidence that these observations are genuine outliers (**Table 7**).

ID	Outcome	Predicted Probability	Deviance Residual	Pearson Residual
10576	Died	0.0178	2.838	7.421
7861	Died	0.0229	2.749	6.535
10813	Died	0.0245	2.723	6.307
22546	Died	0.0306	2.640	5.626
16108	Died	0.0318	2.626	5.520
21249	Died	0.0369	2.569	5.112
7108	Died	0.0392	2.545	4.952
12372	Died	0.0410	2.528	4.837
9906	Died	0.0413	2.525	4.817
23167	Died	0.0419	2.519	4.781

Table 7: Top 10 Outliers by Deviance and Pearson Residuals

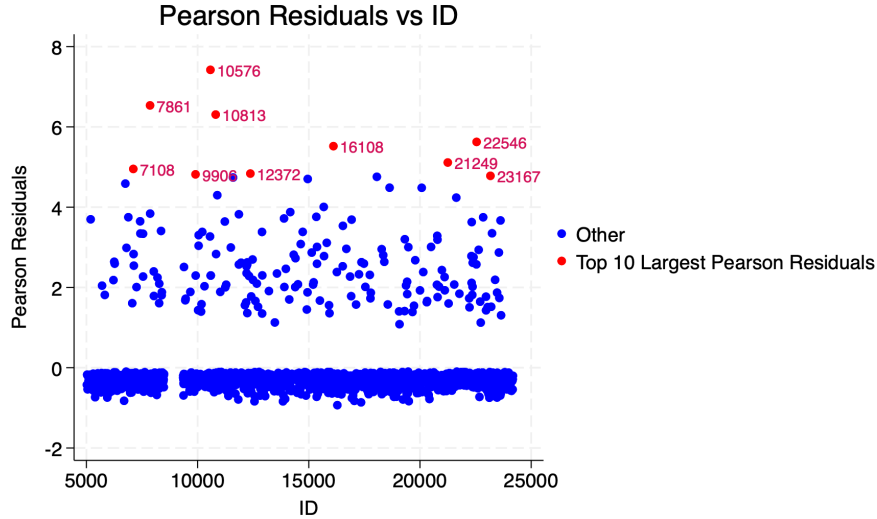


Figure 13: Pearson Residuals vs Observation ID

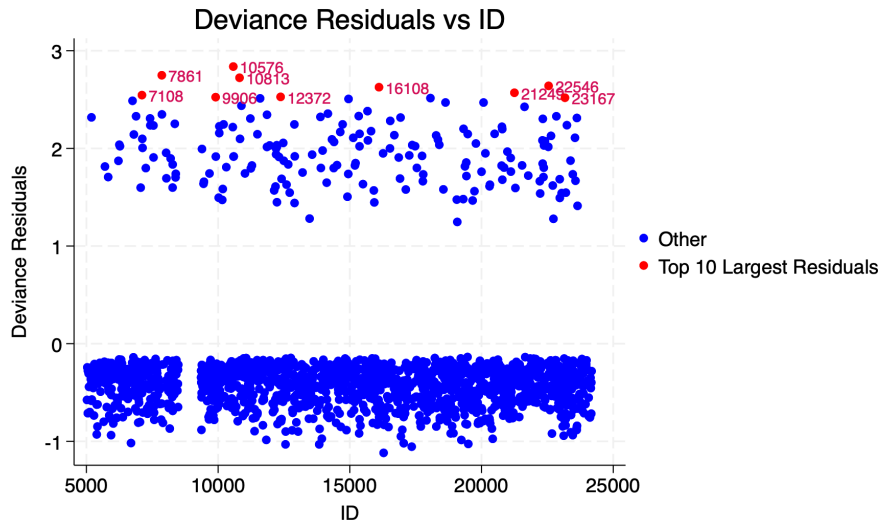


Figure 14: Deviance Residuals vs Observation ID

6.1 What is so weird about these individuals?

A distinct pattern can be seen when the predictor values for the ten outliers are examined (**Table 8**). With ages ranging from 40 to 53 ($\text{age}_{10} = 4.0$ to 5.3), nine of the ten academics are female. Additionally, all ten observations have normal clinical measurements, with BMI, blood pressure, and cholesterol falling within normal ranges. The model was shocked by these deaths because it gives females and younger academics a significantly lower mortality risk. Despite having low risk profiles across all measured predictors, these individuals passed away during the follow-up period, indicating that their deaths may have been caused by unmeasured risk factors not included in the dataset.

ID	Job Grade	Age	Sex	Smoker	SBP	DBP	Chol	BMI
10576	Lecturer	40	Female	Yes	147	86	5.06	25.84
7861	Lecturer	48	Female	No	138	95	7.65	23.33
10813	Post-Doc	48	Female	Yes	143	100	6.85	24.16
22546	Post-Doc	53	Female	No	101	65	4.91	20.22
16108	Lecturer	50	Female	No	144	92	5.68	26.87
21249	Senior Lecturer	49	Female	No	188	103	4.01	24.59
7108	Senior Lecturer	47	Female	Yes	120	75	7.16	23.40
12372	Lecturer	42	Male	No	124	81	4.91	23.24
9906	Senior Lecturer	47	Female	Yes	177	96	8.17	27.02
23167	Lecturer	49	Female	Yes	127	76	6.49	24.80

Table 8: Co-variate Values of the 10 Outliers

7 High Leverage Detection

To identify high-leverage observations, hat values were plotted against observation ID, with a threshold of $2(p+1)/n=0.0127$ indicated by a red dashed line (**Figure 15**). The 10 observations with the highest leverage were identified and highlighted in red. Notably, none of these high-leverage observations were among the top 10 outliers identified in the residual analysis. This suggests that while these cases have unusual co-variate patterns, the model is still able to classify them reasonably well.

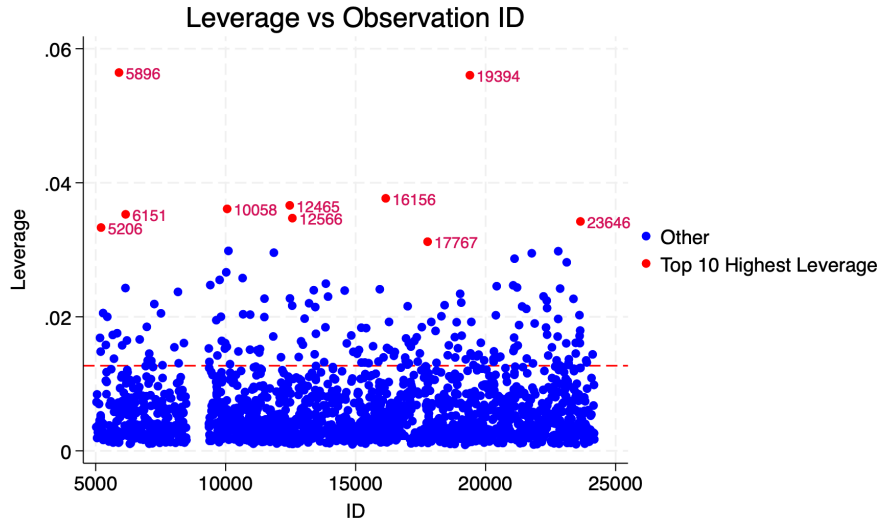


Figure 15: Leverage vs Observation ID

7.1 What causes the high leverage?

A distinct pattern can be seen when the co-variate values of the ten high leverage observations are examined (Table 8). With ages ranging from 55 to 63, all ten observations

are male and among the oldest in the cohort. Furthermore, some observations show abnormally high clinical measurements. The most notable example is observation 16156, which has an exceptionally high systolic blood pressure of 267 mmHg, which explains his high leverage. As previously mentioned, none of these observations were identified as outliers, indicating that despite their peculiar co-variate profiles, the model is still able to predict their results fairly well.

ID	Leverage	Pred. Prob	Job Grade	Age	Sex	Smoker	SBP	DBP	Chol	BMI
5896	0.0564	0.254	Senior Lecturer	62	Male	No	162	93	11.99	21.78
19394	0.0560	0.286	Post-Doc	63	Male	Yes	137	82	6.80	26.35
16156	0.0377	0.178	Lecturer	58	Male	No	267	155	5.04	26.02
12465	0.0366	0.162	Post-Doc	60	Male	No	155	110	4.13	28.08
10058	0.0361	0.144	Post-Doc	58	Male	No	189	109	3.41	23.71
6151	0.0353	0.251	Professor	63	Male	Yes	181	141	5.19	23.49
12566	0.0347	0.333	Senior Lecturer	63	Male	No	100	77	3.72	33.62
23646	0.0342	0.369	Lecturer	61	Male	Yes	201	85	4.50	28.66
5206	0.0333	0.151	Post-Doc	55	Male	Yes	120	69	6.85	24.42
17767	0.0312	0.149	Post-Doc	59	Male	No	148	90	6.98	25.68

Table 9: Co-variate Values of the 10 High Leverage Observations

8 High Influence Detection

To identify influential observations, DBeta values were plotted against observation ID, with the 10 highest values highlighted in red (**Figure 16**). A comparison between these influential observations and the previously identified outliers revealed that four observations—IDs 10813, 22546, 9906, and 21249—appeared in both groups. This indicates that these cases are both poorly predicted by the model and have a substantial impact on the coefficient estimates. Notably, none of the top 10 influential observations overlapped with the high-leverage group.

Conclusion: Given that these four observations are flagged across multiple diagnostic measures, it would be advisable for us to exclude them and refit the model to assess whether their removal improves the model.

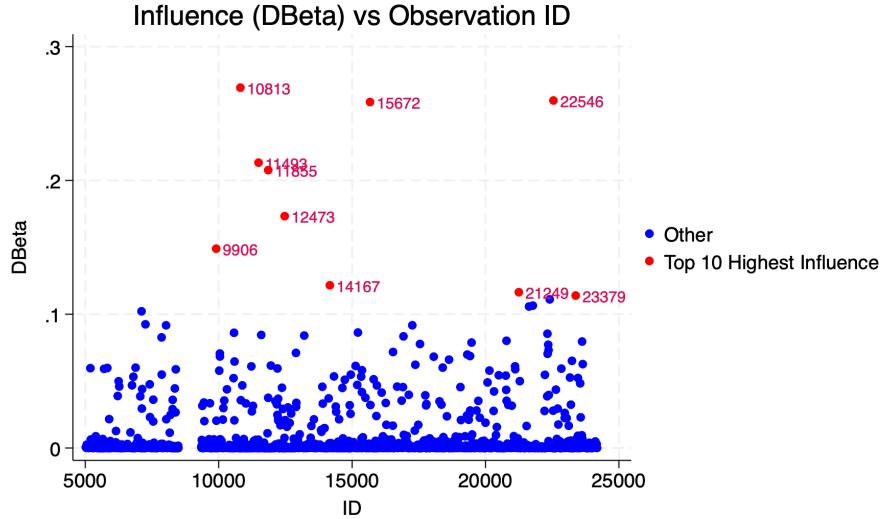


Figure 16: DBeta vs Observation ID

9 External Evaluation of EU Model

9.1 External Calibration of the EU Model

To test how well the EU model worked with Irish data, we applied its original risk formulas to the Irish group. The results, shown in **Figure 17**, reveal a consistent pattern: the model overestimated mortality risk across the board. This is confirmed by an observed-to-expected (O:E) ratio of 0.707, meaning the EU model predicted about 41% more deaths than actually occurred.

The calibration-in-the-large analysis revealed a statistically significant negative CITL of -0.426 ($p < 0.001$), providing further evidence of overestimation of the baseline mortality risk (**Table 10**).

Variable	Coefficient	p-value	95% CI
EU Linear Predictor	1	—	—
Constant	-0.426	0.00	-0.58 — -0.27

Table 10: Calibration in the Large: EU Model Applied to Irish Data

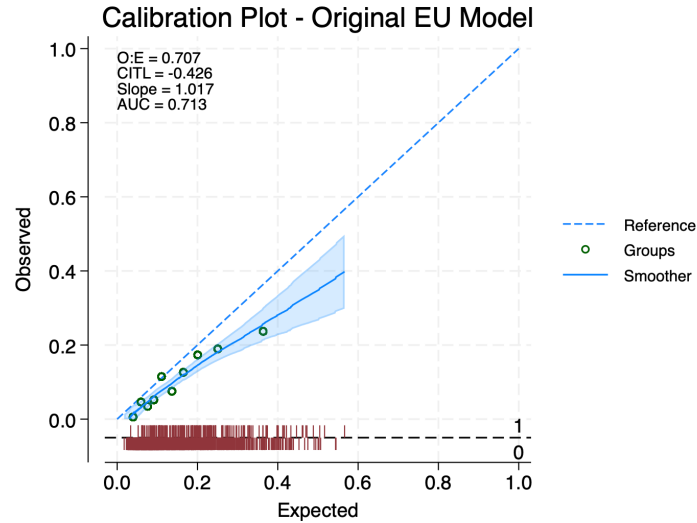


Figure 17: External Calibration Plot of EU Model

9.2 External Discrimination of EU Model

The discriminative ability of the EU model was assessed using the ROC curve (**Figure 18**). The model achieved an AUC of 0.7133, indicating acceptable discrimination, meaning the model correctly identifies the higher risk academic 71.33% of the time. This is slightly lower than the AUC of 0.7354 achieved by the internally fitted Irish model, which is expected given the EU model was developed on a different dataset.

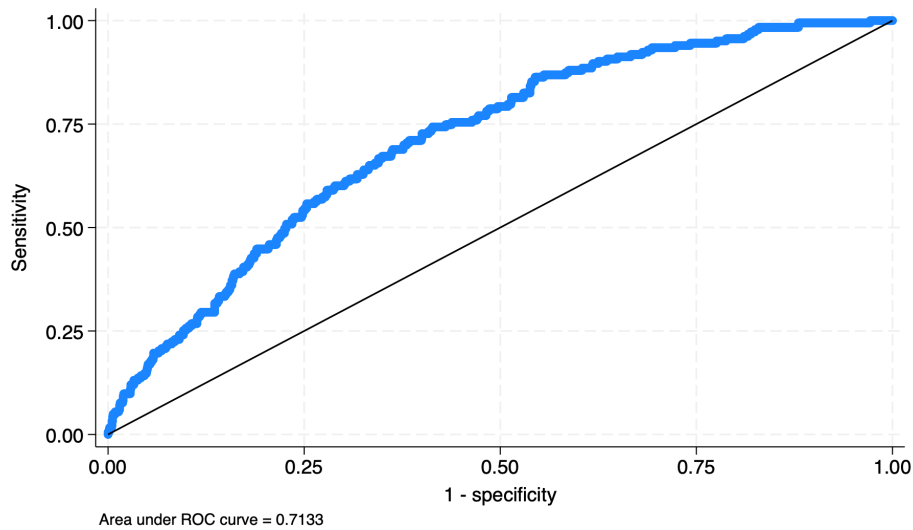


Figure 18: External Discrimination plot of EU Model

10 Recalibration of the EU Model

To recalibrate the EU prediction model for Irish academics, two approaches were considered. First, a full recalibration was performed by freely estimating both the intercept and slope of the EU linear predictor. The calibration slope was 1.017 (95% CI: 0.801–1.233), which is very close to 1. This means the original risk factors in the EU model are already weighted correctly for the Irish population (**Table 11**). Because of this, we only needed to adjust the intercept, keeping the slope fixed at 1.

After recalibration, the model achieved a CITL of 0.000 and an O:E ratio of 1.000, meaning the original overestimation has been fully fixed. In **Figure 19**, the risk groups now fall much closer to the reference line than before—showing clear improvement in calibration. As expected, the AUC stayed at 0.713 because recalibration at the large does not affect discrimination.

Variable	Coefficient	p-value	95% CI
EU Linear Predictor	1.017	<0.001	0.801–1.233
Constant	−0.400	0.029	−0.761–−0.040

Table 11: Full Calibration: EU Model Applied to Irish Data

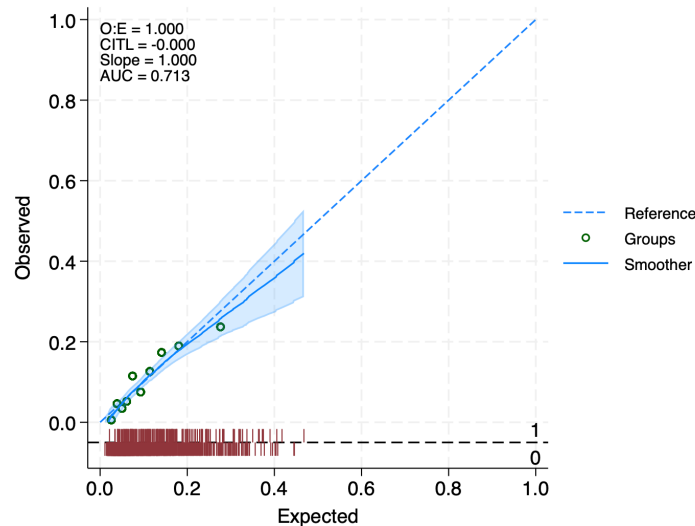


Figure 19: Calibration Plot of EU Model After Intercept Recalibration

Code Used

```
tabstat age sbp dbp chol bmi, stats(mean min max) columns(stats
)
```

```
tab1 jobgrade died smoker sex

twoway (fpfitci died age, estcmd(logistic))
twoway (fpfitci died sbp, estcmd(logistic))
twoway (fpfitci died dbp, estcmd(logistic))
twoway (fpfitci died chol, estcmd(logistic))
twoway (fpfitci died bmi, estcmd(logistic))
graph bar (count), over(died) over(jobgrade) percentages
    asyvars blabel(bar, format(%9.1f)) ytitle("Percentage (%)")
    title("Mortality Rate by Job Grade") legend(label(1 "Survived"
    ") label(2 "Died"))
graph bar (count), over(died) over(smoker) percentages asyvars
    blabel(bar, format(%9.1f)) ytitle("Percentage (%)") title("
    Mortality Rate by Smoking Status") legend(label(1 "Survived")
    label(2 "Died"))
graph bar (count), over(died) over(sex) percentages asyvars
    blabel(bar, format(%9.1f)) ytitle("Percentage (%)") title("
    Mortality Rate by Sex") legend(label(1 "Survived") label(2 "
    Died"))

gen age10 = age / 10
gen sbp10 = sbp / 10
gen dbp10 = dbp / 10
gen chol10 = chol / 10

logistic died i.jobgrade age10 i.sex i.smoker sbp10 dbp10
    chol10 bmi, base

margins jobgrade, at(smoker=0 sex=1) atmeans
marginsplot

predict pdeath, p
estat gof, group(10) table
pmcalplot pdeath died

lroc
```

```
graph hbox pdeath, over(died)

crossfold logistic died i.jobgrade age10 i.sex i.smoker sbp10
    dbp10 chol10 bmi, k(10)

predict dbeta, dbeta
predict dresid, deviance
predict presid, residuals
predict leverage, hat
gen id = _n

* Outliers
gen abs_dresid = abs(dresid)
gsort -abs_dresid
gen top10_res = (_n <= 10)
twoway (scatter dresid id if top10_res == 0, mcolor(blue) msize
    (small)) (scatter dresid id if top10_res == 1, mcolor(red)
    msize(small) mlabel(id)), title("Deviance Residuals vs ID")
    ytitle("Deviance Residuals") xtitle("ID") legend(label(1 "
    Other") label(2 "Top 10 Largest Residuals"))

gen abs_presid = abs(presid)
gsort -abs_presid
gen top10_pres = (_n <= 10)
twoway (scatter presid id if top10_pres == 0, mcolor(blue)
    msize(small)) (scatter presid id if top10_pres == 1, mcolor(
    red) msize(small) mlabel(id)), title("Pearson Residuals vs
    Observation ID") ytitle("Pearson Residuals") xtitle("ID")
    legend(label(1 "Other") label(2 "Top 10 Largest Pearson
    Residuals"))

list id died pdeath dresid presid if id==10576 | id==7861 | id
    ==10813 | id==22546 | id==16108 | id==21249 | id==7108 | id
    ==12372 | id==9906 | id==23167
list id jobgrade age10 sex smoker sbp10 dbp10 chol10 bmi if id
    ==10576 | id==7861 | id==10813 | id==22546 | id==16108 | id
    ==21249 | id==7108 | id==12372 | id==9906 | id==23167

* Leverage
```

```

gsort -leverage
gen top10_lev = (_n <= 10)
twoway (scatter leverage id if top10_lev == 0, mcolor(blue)
       msize(small)) (scatter leverage id if top10_lev == 1, mcolor(
       red) msize(small) mlabel(id)), yline(0.0127, lcolor(red)
       lpattern(dash)) title("Leverage vs Observation ID") ytitle("
       Leverage") xtitle("ID") legend(label(1 "Other") label(2 "Top
       10 Highest Leverage"))
list id leverage pdeath jobgrade age10 sex smoker sbp10 dbp10
     chol10 bmi in 1/10

* Influence
gsort -dbeta
gen top10_db = (_n <= 10)
twoway (scatter dbeta id if top10_db == 0, mcolor(blue) msize(
       small)) (scatter dbeta id if top10_db == 1, mcolor(red) msize
       (small) mlabel(id)), title("Influence (DBeta) vs Observation
       ID") ytitle("DBeta") xtitle("ID") legend(label(1 "Other")
       label(2 "Top 10 Highest Influence"))
list id dbeta if top10_db == 1 & (id == 10576 | id == 7861 | id
     == 10813 | id == 22546 | id == 16108 | id == 21249 | id ==
     7108 | id == 12372 | id == 9906 | id == 23167)

gen eu_lp = -7.9048 + 0.3639*(jobgrade==2) + 0.5344*(jobgrade
     ==3) + 0.6438*(jobgrade==4) + 0.8180*(sex==2) + 0.5663*(
     smoker==2) + 0.0108*sbp + 0.0631*age + 0.0054*bmi
gen eu_phat = invlogit(eu_lp)
pmcalplot eu_phat died, title("Calibration Plot - Original EU
     Model")
roctab died eu_phat, graph title("ROC Curve - Original EU Model
     ")

logistic died eu_lp, coef
constraint 1 eu_lp = 1
logistic died eu_lp, coef constraint(1)
predict recal_phat, p
pmcalplot recal_phat died, title("Calibration Plot -
     Recalibrated Model")

```